

Forensic Intelligence Graphs: An LLM Approach to Digital Evidence Extraction and Relationship Analysis

Honghe Zhou*, Weifeng Xu[†], Josh Dehlinger*, Suranjan Chakraborty*, and Lin Deng*

* Department of Computer and Information Sciences

Towson University, Maryland, USA

Emails: hzhou4@students.towson.edu, {jdehlinger, schakraborty, ldeng}@towson.edu

[†]School of Criminal Justice

University of Baltimore, Maryland, USA

Email: wxu@ubalt.edu

Abstract—Digital forensics often requires deriving meaningful and investigative intelligence from vast amounts of evidence scattered across various artifacts. In this study, we propose an automated approach to gain insights about criminal incidents using digital evidence networks constructed with the aid of Large Language Models (LLMs). Our method utilizes LLMs to extract evidence entities from mobile devices and infers relationships among them. Using this information, the model enables the generation of Forensic Intelligence Graphs (FIGs). These graphs visually represent evidence entities and their interrelations, providing an intelligence-driven approach to forensic data analysis. Using evidence extracted from Android mobile devices, an empirical evaluation demonstrates that the LLM-aided FIG achieves 93.33% coverage of evidence entities and 86.96% coverage of evidence relationships, effectively uncovering all relevant suspect scenarios. Moreover, our approach uncovered 27 additional evidence entities and 83 relationships beyond those recorded in the official documentation, highlighting its ability to reveal previously overlooked forensic artifacts.

Index Terms—Digital Forensics, Large Language Models, Forensic Intelligence Graphs

I. INTRODUCTION

The exponential growth of digital devices and the proliferation of online interactions have transformed the way evidence is generated, stored, and analyzed, presenting both opportunities and challenges for digital forensics. With billions of users relying on digital platforms for communication, commerce, and social networking, a vast amount of forensic artifacts, such as names, addresses, phone numbers, email addresses, and usernames, are embedded within unstructured and dispersed data sources. By 2024, the number of smartphone users surpassed 7.21 billion globally, significantly contributing to the surge in digital evidence generated in diverse environments [1].

Digital forensics is crucial in combating cybercrime, yet investigators face substantial hurdles in managing these vast and complex data. Traditional approaches to extracting and analyzing evidence artifacts in digital forensics are fraught with limitations, primarily due to their reliance on manual processes and predefined rules [2]. These methods struggle to keep up with the complexity and diversity of data formats

across modern devices and applications [3]. For example, evidence such as usernames or email addresses may be stored in various formats, ranging from structured fields in databases to unstructured text within chat logs or application metadata. Furthermore, the sheer volume of data often requires labor-intensive efforts to identify, correlate, and analyze evidence, making it difficult to derive meaningful and investigative insights. This inefficiency not only slows down investigations but also increases the risk of missing critical connections between evidence artifacts. As the sophistication of cybercrime continues to evolve, there is a pressing need for advanced techniques capable of automating the extraction and analysis of forensic artifacts while managing the inherent variability and scale of digital evidence. To address these challenges, this research leverages Large Language Models (LLMs) to enhance evidence artifact extraction and construct Forensic Intelligence Graphs (FIGs). Using the advanced capabilities of LLM, we aim to improve the precision, efficiency, and comprehensiveness of digital forensic investigations, ultimately enabling investigators to uncover complex relationships and derive meaningful and investigative criminal insights more effectively.

Our approach is structured into four primary steps to address the complexities of digital forensic investigations. The first step involves converting the raw data into standardized plain text. This process begins with extracting data from Embedded MultiMediaCards (eMMCs) on mobile devices, which often store potential evidence entities in binary formats, such as deleted, hidden, and fragmented data. Transforming the raw binary data into plain text enables LLMs to effectively identify and analyze evidence artifacts. The second step focuses on discovering evidence entities and their interconnections by designing and refining customized LLM prompts. These prompts guide the LLM to analyze text files line by line, efficiently extracting relevant artifacts, and identifying relationships between them. The third step focuses on inferring the relationships between evidence entities and constructing evidence networks. Using outputs generated by LLMs and

the original storage structure of the files, connections are established based on the degree of direct or indirect linkage and the number of intermediary points in indirect connections. This information is then used to automatically generate FIGs, which visually represent the relationships among evidence entities, offering a robust method for analyzing forensic data and uncovering complex interrelations. Finally, the fourth step focuses on deriving insights into criminal behaviors. By analyzing the automatically generated FIGs, critical conclusions about criminal patterns, relationships, and activities are drawn. These insights enable investigators to efficiently uncover meaningful and investigative intelligence, providing a comprehensive understanding of criminal behaviors and their context.

Through an experimental evaluation using an open-source digital forensic dataset, we demonstrate that our approach achieves an Evidence Entity Coverage of 93.33%, successfully identifying 42 out of 45 documented evidence entities, and an Evidence Relationship Coverage of 86.96%, correctly inferring 20 out of 23 documented relationships.

Moreover, our approach uncovered 27 additional evidence entities and 83 additional relationships beyond those recorded in the official documentation of the dataset, showcasing its ability to identify previously overlooked forensic artifacts.

The contributions of this paper are as follows:

- We propose a novel approach for automating the extraction and analysis of forensic evidence, addressing the limitations of traditional manual techniques.
- We leverage LLMs to extract evidence artifacts, such as names, addresses, phone numbers, emails, and usernames, from unstructured text data, ensuring accurate and efficient artifact identification.
- We design and implement an innovative approach to construct evidence networks by linking isolated evidence entities based on their direct and indirect relationships. This approach results in the creation of FIGs, which visually represents interconnected evidence entities and provide investigators with meaningful and investigative insights into criminal patterns and behaviors.
- We release an open-source toolset¹ to facilitate further research and development of LLM-based digital forensic analysis methods.

This paper is organized as follows. Section II provides the background of digital forensics, mobile forensics, LLMs, and fundamental concepts of our research. Section III discusses related work in digital forensics on mobile devices. Section IV details our proposed approach. Section V introduces our experiment on real-world Android mobile device image datasets and evaluation results. Section VI discusses the challenges in our research and provides our solutions. Section VII provides threats to validity. Finally, Section VIII provides concluding remarks.

¹<https://github.com/HongheZ/An-LLM-driven-Approach-to-Gain-Cybercrime-Insights-with-Evidence-Networks>

II. BACKGROUND

A. Digital Forensics and Mobile Forensics

Digital forensics refers to the application of computer science and investigative techniques to uncover, analyze, and interpret digital evidence for legal and investigative purposes. It is a specialized branch of traditional forensic science [3], [4]. Digital evidence, integral to many modern investigations, is distributed across a wide variety of artifacts, including operating system registries, device memory, service logs, emails, and other sources. The digital forensics investigation lifecycle encompasses several critical stages: identification, collection, recovery, analysis, and preservation of digital evidence [5].

Mobile forensics is a vital sub-discipline of digital forensics that involves retrieving, analyzing, and presenting digital evidence from mobile computing devices such as smartphones and tablets in a forensically sound and legally admissible manner. The widespread use of mobile devices, particularly Android and iOS, has made them significant sources of evidence in criminal investigations and primary targets for crimes. Mobile forensics encompasses tasks like recovering deleted data, analyzing call logs, and tracing application activities while addressing challenges such as data fragmentation, encryption, device heterogeneity, and diverse operating systems. Techniques range from manual extraction to advanced methods like chip-off and micro-reading, ensuring evidence integrity despite rapid technological advancements and sophisticated security features. As cyber threats and device capabilities evolve, continuous innovation in tools and methods is essential to maintain the effectiveness of mobile forensics in modern digital investigations [6], [7].

B. Large Language Models (LLMs)

Large Language Models (LLMs) represent a significant advancement in natural language processing (NLP), characterized by their extensive parameter sizes and exceptional learning capabilities [8], [9]. These models are built upon the Transformer architecture, which introduced the revolutionary self-attention mechanism. The Transformer has reshaped the field of NLP by efficiently handling sequential data and capturing long-range dependencies within text [10]. Its parallelization capabilities have further enhanced the scalability and performance of LLMs, enabling them to process and analyze complex language tasks with remarkable accuracy.

Another defining feature of LLMs is in-context learning, which allows the model to generate text based on a provided context or prompt [11]. This capability enables LLMs to produce coherent and contextually relevant responses, making them ideal for various applications, including conversational agents, content generation, and interactive systems [12]. To enhance forensic analysis, our approach leverages powerful NLP capabilities and in-context learning, which are two key features of LLMs. By utilizing their advanced NLP abilities, LLMs can accurately extract evidence entities from diverse and unstructured data sources, ensuring precise identification of various types of evidence entities. Additionally, through in-context learning, we can define inclusion criteria to specify the

characteristics that a target evidence entity should possess and exclusion criteria to filter out irrelevant entities. This process enables our approach to extract the most relevant forensic artifacts with greater accuracy and reliability.

C. Fundamental Concepts

1) *Evidence Entity*: Evidence entities represent fundamental components of digital evidence in forensic investigations. We define $E = \{e_1, e_2, \dots, e_n\}$ as the set of evidence entities, where each entity e is defined as: $e = (Content, Type)$ where *Content* refers to the specific information or data associated with the evidence, and *Type* denotes the classification of this evidence, identifying its nature and purpose within the context of an investigation, such as a person's name, address, or phone number. For example, an entity with the evidence content "Josh Hickman" and the evidence type "Person's Name" can be represented as: $e = ("Josh Hickman", \text{Person's Name})$.

2) *Evidence Relationship*: Evidence relationships represent the connections between two distinct evidence entities in digital forensic investigations. We define $R = \{r_1, r_2, \dots, r_m\}$ as the set of relationships between evidence entities. Each relationship r between two evidence entities e_i and e_j is defined as: $r = (e_i, connection, e_j)$ where $e_i, e_j \in E$ and *connection* describes the nature of the connection between these entities.

Examples of such connection include: "own": indicating ownership, e.g., a person owns a phone number, username, or other evidence entity. "live-in": indicating residency, e.g., a person lives in an address. If a person named "Josh Hickman" owns the username "thisisdfr," this evidence relationship can be represented as $r = (\text{Josh Hickman}, \text{own}, \text{thisisdfr})$.

3) *Forensic Intelligence Graphs (FIGs)*: Forensic Intelligence Graphs (FIGs) are a method for representing digital forensic evidence using knowledge graphs [13]. A knowledge graph is a structured representation of entities and their relationships, designed to organize information and facilitate reasoning. It captures real-world objects or abstract concepts as nodes and connects them with labeled edges that describe the relationships. For example, in digital forensics, a knowledge graph can link evidence, actions, and timelines, enabling investigators to analyze and present complex data in an understandable and contextually rich manner. A FIG is defined as $G = (E, R)$, where E is the set of evidence entities, represented as nodes in the graph. And R is the set of evidence relationships, represented as edges in the graph.

Fig. 1 shows a FIG example generated by our proposed approach. The graph comprises three nodes, representing a person's name, a telephone number, and an email address. The edges labeled "own" denote ownership relationships, indicating that the person, "Josh Hickman," owns the telephone number "9195794674" and the email address "thisisdfr@gmail.com."

III. RELATED WORK

Researchers have shown significant interest in exploring methodologies to enhance the efficiency and accuracy of digital forensics on mobile devices. A variety of techniques have

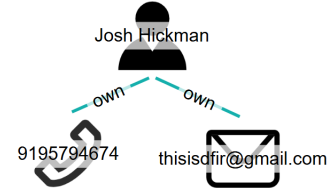


Fig. 1. A FIG Example

been proposed, ranging from traditional rule-based approaches and manual data extraction methods to advanced machine learning models and automated analysis frameworks.

A. Traditional Mobile Forensics Methods

Traditional methods for mobile forensics primarily rely on manual processes and predefined rules for data extraction and analysis. These methods include logical and physical acquisition techniques, as well as analysis through predefined patterns. For example, Roy *et al.* [14] provides a comprehensive survey of methods and tools used in Android mobile forensic analysis, including manual, physical, and logical data acquisition techniques. It highlights the challenges posed by diverse Android device architectures and evolving software, which complicate the standardization of forensic tools. Similarly, Sathe *et al.* [15] compared commercial and open-source forensic tools for logical and physical data extraction. The results highlight that no single tool is comprehensive enough, and investigators should employ multiple tools to ensure effective forensic analysis. Moreover, Lwin *et al.* [16] highlights that traditional methods encounter significant challenges due to the growing diversity of data formats. Furthermore, command-line tools such as ADB Backup and DD require specialized technical expertise, and no single tool is capable of comprehensively acquiring and analyzing all types of data. Consequently, manual analysis becomes not only time-intensive but also prone to errors.

B. Machine Learning-Based Mobile Forensic Methods

To overcome the limitations of traditional approaches, researchers have explored machine learning techniques for forensic investigations. For instance, Peng *et al.* [17] propose a machine learning-based framework to enhance mobile forensics by analyzing documents and images from smart devices. The framework employs kernel k-means clustering to handle non-linear separable clusters in document analysis and multi-class SVM to classify image data efficiently. Additionally, Marturana *et al.* [18] presents a quantitative approach to mobile forensic triaging, focusing on using machine learning algorithms to classify mobile phones based on their likelihood of involvement in crimes like pedophilia. The authors propose a workflow that incorporates data normalization, feature extraction, and classification using tools like WEKA, with algorithms including Bayesian Networks, Decision Trees, and Locally Weighted Learning (LWL).

However, machine learning methods in mobile forensics continue to face significant challenges, particularly due to the

diversity of mobile devices and tool compatibility issues [19]. The wide range of hardware and software platforms, combined with the rapid evolution of device models, complicates the development of universal forensic tools. Existing tools often struggle to maintain compatibility across various device types and operating systems, resulting in incomplete or inaccessible data extraction. Furthermore, machine learning models trained on specific scenarios may fail to capture the complexity of real-world forensic cases, leading to a lack of generalizability and potential overfitting [18].

Our research aims to address these challenges by developing an automated model that leverages LLMs to enhance mobile forensic processes.

IV. PROPOSED APPROACH

This section introduces our approach to leveraging LLMs to extract evidence entities, and identify relationships, and FIGs for digital forensic investigations. In this approach, we selected the gpt-4o model as the core LLM. The gpt-4o model is the latest advancement from OpenAI and a highly sophisticated language model designed to understand and generate human-like text with exceptional precision and contextual awareness [20]. This model excels in tasks such as interactive dialogue, content generation, and complex text analysis, making it particularly well-suited for specific content extraction in digital forensics.

The core concept of our approach is to preprocess raw forensic data, including unstructured text and binary files, and combine it with carefully designed prompts tailored for forensic tasks. The preprocessed data are then used in conjunction with the gpt-4o model to extract evidence entities, determine their relationships, and construct FIGs that visually represent interconnected evidence. Fig. 2 illustrates our general approach.

A. Phase 1: Data Collection and Preprocessing

The first phase of our approach involves examining and processing data stored on mobile devices' Embedded MultiMediaCards (eMMCs). eMMCs are critical sources of potential evidence, containing entities such as deleted, hidden, and fragmented data, which are stored in binary format. To facilitate further analysis, raw data from eMMCs is extracted and converted into standardized text. During preprocessing, redundant, corrupted, or irrelevant files (e.g., cache files, duplicate files, thumbnail images) are filtered out to retain only the most pertinent evidence. Files that typically contain user information are prioritized, such as database files, log files, operating system files, application settings files, and so on.

B. Phase 2: Prompt Design and Evidence Extraction

In the second phase, we combine the preprocessed data, formatted as standardized text files, with well-designed prompts to enable LLMs to effectively analyze and extract evidence artifacts. Various prompts are developed and tailored specifically for the forensic analysis of this preprocessed data. Prompt engineering plays a critical role in this phase, where the

prompts are iteratively refined to maximize their effectiveness and guide the LLMs in delivering optimal results. After the prompt design is finished, another key aspect of this phase is the discovery of evidence entities and their interconnections. The refined prompts are applied to the standardized text files containing the preprocessed data and analyzed line by line, enabling the efficient extraction of relevant evidence entities. The detailed algorithm of prompt construction and prompt template is shown as algorithm 1.

Algorithm 1: Prompt Construction

Data: Preprocessed Dataset, Target Evidence Entity Type
Result: Constructed Prompts

```

1 for each row  $r$  in Preprocessed Dataset do
2   Prompt Template  $\leftarrow$  "You are an expert text
   analyzer specializing in extracting <Target
   Evidence Entity Type> from text. The
   additional rule you should follow for this kind of
   evidence entity is <Inclusion criteria
   for the specific type of evidence
   entity> If you find any <Target Evidence
   Entity Type>, output each one you found line
   by line. If you can't find any <Target
   Evidence Entity Type> in the text, only
   output 'None'. Before outputting, ensure that each
   name is <Exclusion criteria for the
   specific type of evidence entity>
   Here is the text line:  $r$ "
3 end
4 Replace <Target Evidence Entity Type>
   with desire type of the evidence entity;
5 Replace <Inclusion criteria for the
   specific type of evidence entity> with
   the characteristics that the extracted result should
   have;
6 Replace <Exclusion criteria for the
   specific type of evidence entity> with
   the characteristics that the extracted result should not
   have;
7 Construct prompt  $p$  using the filled template;
8 return Constructed Prompts
```

In this algorithm, the <Target Evidence Entity Type> refers to the desired type of evidence entity specified by the user. This may include various types of personal information, such as a person's profile, which can contain their name, username, address, email, telephone number, and similar details. The <Inclusion criteria for the specific type of evidence entity> and <Exclusion criteria for the specific type of evidence entity> define the rules that govern the characteristics the extracted result should possess and avoid, respectively. These criteria are incorporated into the prompt to enhance the accuracy and reliability of the LLM's output. For example, if the <Target Evidence Entity

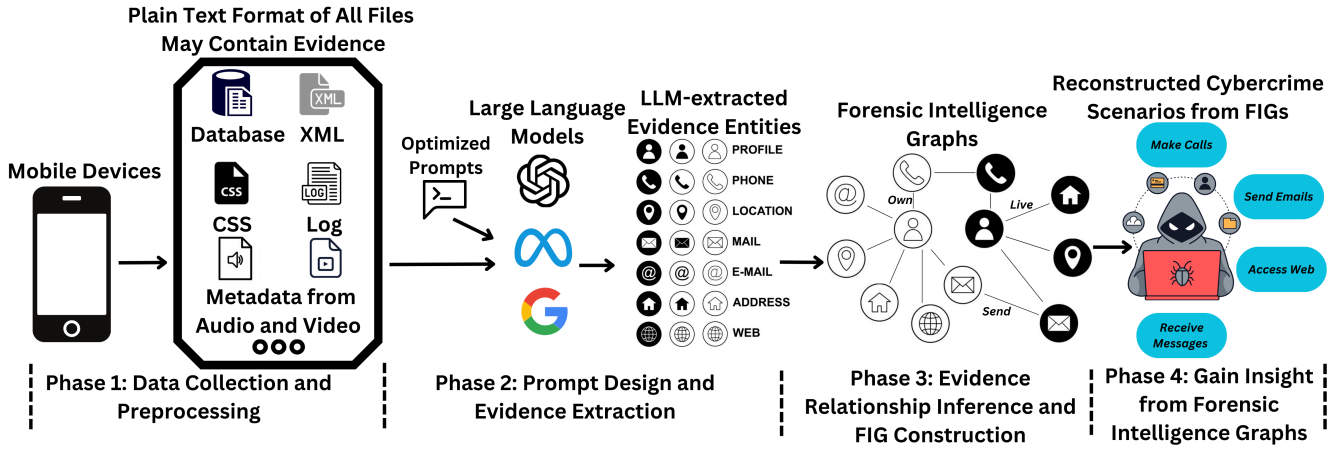


Fig. 2. An LLM-Driven Approach for Gaining Cybercrime Insights with Forensic Intelligence Graphs

Type> is “person’s name,” the prompts must ensure that the LLM extracts only real names and not usernames or nicknames. Additionally, certain real names may contain special characters, such as underscores or hyphens, which must be preserved in their original format during extraction. In this case, the <Inclusion criteria for the specific type of evidence entity> would be: “You should only extract real names. Preserve the original format of the names exactly as they appear in the input, including underscores or other characters.” Meanwhile, the <Exclusion criteria for the specific type of evidence entity> would specify: “Ensure that each name is a real person’s name and not a username or nickname.” This example prompt is provided below:

User input prompt: You are an expert text analyzer that specializes in extracting the person’s name from text. You should only extract real names. Preserve the original format of the names exactly as they appear in the input, including underscores or other characters. If you find any person’s names, output each one you found line by line. If you can’t find any person’s name in the text, only output ‘None’. Before outputting, ensure that each name is a real person’s name and not a username or nickname. Here is the text line: <Current text row>

Output format:

```
<Preson’s name1>
<Preson’s name2>
<Preson’s name3>
...
```

C. Phase 3: Evidence Relationship Inference and FIG Construction

In this step, we design an approach to infer relationships between evidence entities by analyzing their data storage patterns and construct evidence networks to visualize these connections. Data storage patterns refers to the logical relative closeness or distance of data storage among different

evidence entities. This is used as a key factor for establishing relationships.

To analyze the data storage patterns, our approach collects data location metadata for each evidence entity. Data location metadata refers to comprehensive information regarding an entity’s storage position. The data location metadata for different file types corresponds to the specific structural details of the file type in which an evidence entity exists. For instance, for an evidence entity in a database file, its data location metadata includes the row number, the column name, the table name, and the database name where the entity is found. For an evidence entity in an XML file, its data location metadata includes the line number and tag name where the evidence entity appears, along with the parent, sibling, and child tag names associated with it.

Then, according to evidence entities’ data location metadata, our approach defines different classes for the data storage patterns. Different classes will lead to significantly different FIGs. The following lists the primary classes of data storage patterns:

- **Same Device:** Two entities are extracted from different applications, indicating that they coexist on the device but do not share an application-specific relationship.
- **Same Application:** Two entities belong to the same application but are stored in different files, suggesting they are part of the same application ecosystem but may not be directly connected.

Furthermore, for evidence entities stored within databases, the data storage patterns are categorized as follows:

- **Same Database:** Both entities reside within the same database.
- **Same Database Table:** Both entities are located in the same table within a database.
- **Adjacent Database Rows:** Two entities are stored in consecutive rows of the same database table.
- **Same Database Row:** Two entities share the same row number, indicating they are part of the same record.

For evidence entities stored within XML files, the data storage patterns are defined as follows:

- **Same XML File:** Both entities are located in the same XML file.
- **Sibling:** Two entities exist at the same hierarchical level under a common parent element.
- **Parent-Child:** One entity is directly nested within another, indicating a structural dependency.
- **Same XML Element:** Both entities reside within the same XML element, meaning they are part of the same structured data unit.

Using the identified evidence entities along with their data location metadata, after determining specific data storage patterns, relationships are established between evidence entities based on their degree of direct or indirect linkage, as well as the number of intermediary points in indirect connections. Our approach infers the relationships between evidence entities by analyzing various contextual factors, including the type of evidence, the application from which the evidence entity was extracted, the number of occurrences of the same evidence entity, and other relevant attributes specific to the file type. For example, consider the relationship between the person’s name *Wesley Skinner* and the username *wesleyskinner7*, both of which were extracted from the Snapchat application. These evidence entities are stored within the same database “main.db,” and occur in two separate tables following the “Same Database Row” storage pattern. In the “Friend” table, both evidence entities are found in row 27, where *Wesley Skinner* is stored under the column “displayName”, and *wesleyskinner7* is stored under the column “username”. Similarly, in another “FriendDeltaForce” table, both entities appear in row 58, with *Wesley Skinner* again recorded under “displayName”, and *wesleyskinner7* under “username”. Given this consistent co-occurrence in multiple tables under the same row, our approach infers that *Wesley Skinner* owns the username *wesleyskinner7*, establishing an ownership relationship between the two entities. This information is then utilized to automatically generate FIGs. FIGs visually represent the relationships among evidence entities, providing a structured and intuitive evidence network. This approach offers a robust method for analyzing forensic data, uncovering complex interrelations, and deriving meaningful and investigative insights for investigations.

D. Phase 4: Gain Insight from Forensic Intelligence Graphs

In this phase, we focus on deriving meaningful and investigative insights into criminal behaviors by analyzing the automatically generated FIGs. These graphs, which visually represent evidence entities and their relationships, serve as a structured framework for uncovering complex patterns and connections within forensic data.

By examining the FIGs, investigators can identify critical conclusions about criminal patterns, relationships, and activities. For instance, the analysis may reveal connections between individuals, locations, and communication methods, highlighting previously unknown links that are vital to the investigation.

Additionally, FIGs allow for the detection of anomalies and recurring behavioral patterns, providing a deeper understanding of the suspects’ actions and strategies.

The structured visualization provided by FIGs not only enhances the interpretability of forensic data but also facilitates decision-making by offering a clear and intuitive representation of complex evidence networks. This approach empowers forensic professionals to address the intricacies of modern investigations with increased accuracy and efficiency.

V. EXPERIMENTAL EVALUATION

We conduct an experimental evaluation using real-world forensic datasets to assess the effectiveness of our proposed approach. This evaluation aims to address the following three research questions (RQs):

RQ1: Can our proposed approach accurately identify and extract evidence entities from forensic datasets?

RQ2: Can our proposed approach accurately identify evidence relationships from forensic datasets?

RQ3: Can our proposed approach effectively uncover additional evidence entities and relationships beyond those documented in official records?

RQ4: Can the FIG generated by our approach effectively infer meaningful and investigative insights to aid in forensic analysis?

A. Experimental Design

This section introduces the design of our experimental evaluation. This experiment includes the following three steps.

Step 1: Prepare and Preprocess the Experiment Data

The data we used to evaluate our approach is an open-source Android 10 device image created for digital investigations [21]. This image was generated using a stock Android image from Google and populated with real-world user data in both system and third-party apps, simulating authentic usage scenarios to ensure realistic forensic conditions. The Android device image used in our experiment has a total size of 4.5 GB and contains 36,236 files. For our experiment, we focused only on database files (418 in total) and XML files (1084 in total) to keep the dataset manageable and relevant. A database file is a structured collection of data organized to facilitate efficient storage, retrieval, and management of information. In Android devices, applications commonly use SQLite databases to store structured data locally. SQLite database files are commonly used to store important user information, such as contacts, messages, and application data, ensuring that the data is readily accessible and can be managed effectively [22]. XML file is another common file type that stores important user information on mobile devices. The hierarchical nature of XML allows for different views of conceptual information, enabling the nesting of related entities within the same file. This nesting reflects the relationships between entities, ensuring that closely related data is stored together, which can enhance data retrieval and integrity [23].

Before converting these files to a text format suitable for LLM analysis, it is necessary to filter out corrupted and

redundant files. For instance, when accessing database files on an Android device, some files with “.wal” and “.shm” extensions may be present. The “.wal” file is used by SQLite to record database changes before committing them to the main database, enhancing performance and supporting atomic commits and rollbacks. The “.shm” file, on the other hand, serves as an index to the “.wal” file, enabling efficient data sharing among multiple connections and facilitating faster access and coordination between processes [24]. These files are excluded from analysis as they contain duplicate data from the original file.

For database files, we divide the data into rows, treating each row as a separate unit for analysis. Since each row represents a distinct record with related data fields, this approach helps preserve context and allows for a thorough examination of individual evidence entities. For XML files, we utilize their hierarchical tree structure to partition the data. Specifically, we segment XML data at the lowest-level parent nodes in the hierarchy. These are elements that have children but whose children do not have further descendants. These nodes can also be described as the parents of leaf elements. XML files are designed to represent structured data, where each parent node groups related child elements. This partitioning strategy ensures that the closest relationships between evidence entities are preserved and detected accurately during analysis. After preprocessing, the dataset is set up with the designed prompts and ready for the LLM to analyze.

Step 2: Extract Evidence Entities and Construct FIGs In this step, we send the prepared dataset to the LLM for analysis to extract evidence entities and record relevant data location metadata for each entity. During this process, to guarantee the accuracy and reliability of the extracted evidence entities result, we fine-tune the parameters of the LLM delicately to maximize its performance. One of the key adjustments was lowering the *temperature* parameter to its minimum feasible value while ensuring optimal result. Decreasing the temperature guarantees that the LLM will always choose the most likely response, minimizing variability between runs and enhancing reproducibility [25]. Another crucial parameter we fine-tune is *top-k*. *Top-k* limits the model’s choice to tokens with a top k highest probability, lowering the likelihood of choosing low-probability outputs [25].

After extracting the evidence, we identify relationships between related entities based on data storage patterns, forming an initial evidence network. Linking evidence entities based on data storage patterns plays a key role in accurate relationship inference. In this experiment, we decide to employ the data storage pattern of “Same Database Row” for database files and the “Sibling” storage pattern for XML files to ensure a structured and meaningful representation in constructing relationships among entities. This is based on several considerations. In database files, entities within the same row are inherently linked, representing a single record that encapsulates interconnected attributes. This ensures that only strongly related evidence entities are connected, reducing the risk of false positives in the analysis. Similarly, in XML files,

child elements sharing the same parent represent the closest and most contextually meaningful relationships, preserving the logical hierarchy and improving the accuracy of inferred connections. The total number of occurrences of each evidence entity across different applications is also recorded.

In this experiment, we focus on the five most frequently recorded types of evidence entities as specified in the official documentation of the open-source Android image: person’s name, address, phone number, email, and username. For example, if a specific person’s name is associated with a phone number that appears in multiple apps, it can be inferred that the individual owns the phone number. Conversely, if the phone number appears in only one application, it suggests that the individual has interacted with another person who owns the phone number through that application. This inference is based on the common practice of phone numbers being used as a unique identifier for user registration across various apps. Our inference process identifies common relationships between evidence entities across different apps, offering a comprehensive view of their connections. After the inference is completed, our approach utilizes all the recorded information to automatically generate FIGs.

Step 3: Evaluate the Performance The extracted entities and inferred relationships are compared against the official documentation to determine the effectiveness of our method. The primary metrics used to evaluate performance include *Evidence Entity Coverage*, *Evidence Relationship Coverage*, *Additional Evidence Entities Detected*, and *Additional Evidence Relationships Detected*.

Evidence Entity Coverage measures the proportion of correctly detected evidence entities relative to the total number of evidence entities recorded in the official documentation. Specifically, it is defined as:

$$Evidence\ Entity\ Coverage = \frac{|E_{Detected}|}{|E_{Official}|} \times 100\% \quad (1)$$

where $E_{Detected}$ represents the set of correctly detected evidence entities that match those in the official documentation, and $E_{Official}$ denotes the set of evidence entities recorded in the official documentation.

Evidence Relationship Coverage evaluates the proportion of correctly identified relationships between evidence entities compared to those recorded in the official documentation. It is calculated as follows:

$$Evidence\ Relationship\ Coverage = \frac{|R_{Detected}|}{|R_{Official}|} \times 100\% \quad (2)$$

where $R_{Detected}$ indicates the set of correctly detected relationships found in the official documentation, and $R_{Official}$ refers to the set of relationships documented officially.

Additional Evidence Entities Detected quantifies the number of correct evidence entities discovered by our approach that were not listed in the official documentation.

Additional Evidence Relationships Detected measures the number of correct relationships inferred by our approach beyond those recorded in the official documentation.

B. Experimental Results and Discussion

This section shows the experimental results.

1) *Results for RQ1 and RQ2*: We compare the evidence entities identified by our approach with the ground truth provided in the official documentation of the forensic dataset, which records 45 evidence entities and 23 relationships. To ensure the accuracy and reliability of our experimental results, we conduct multiple exhaustive searches across the entire Android device image using commands such as *find* and *grep* to verify the presence of each documented evidence entity. Through this verification process, we identify several entities that were recorded in the official documentation but do not actually exist within the forensic dataset. To prevent inaccuracies in the evaluation, we remove these non-existent entities from the ground truth.

Table I summarizes the experimental results. Our approach achieved an Evidence Entity Coverage of 93.33%, successfully identifying 42 out of 45 evidence entities recorded in the official documentation.

We conducted further analysis on the missing evidence entities to understand the reasons behind their omissions. Specifically, two entities were found in C# source code files, which were outside our experimental scope since we focused on database and XML files. In a follow-up experiment on the C# file, our method successfully extracted these two entities and their relationships, showing that our approach can handle additional file formats. The last missing evidence entity was a phone number in a non-U.S. format. Since our experiment focused on U.S. phone numbers, the LLM prompt excluded other formats, so this entity was intentionally left out.

Our method identified 20 of 23 relationships documented in the official records, achieving an Evidence Relationship Coverage of 86.96%. Analysis of the missing relationships showed they all resulted from missing evidence entities, caused by unavailable or excluded data.

Overall, these findings demonstrate the high precision of our approach in extracting evidence entities and inferring relationships.

2) *Results for RQ3*: As shown in Table I, our method identified 27 additional evidence entities and 83 additional relationships that were not recorded in the official documentation. A comprehensive examination of these additional findings was conducted to verify their relevance and accuracy. The analysis confirmed that all are valid and valuable forensic artifacts. The newly discovered evidence provides deeper insights into user interactions, device usage patterns, and behavioral habits, thereby enhancing the overall comprehensiveness of the forensic analysis. These findings demonstrate the capability of our approach to uncover critical forensic artifacts that might otherwise remain undetected, offering investigators a more comprehensive view of the digital environment under investigation.

3) *Results for RQ4*: Fig. 3 shows the LLM-constructed FIG of our experiment. The FIG contains four central evidence entity nodes with the highest degree. In graph theory, a higher degree indicates a greater number of relationships within the entire Android device, suggesting a higher likelihood of these entities being critical to the investigation.

The first central evidence entity identified is “thisisd-fir@gmail.com”, which is documented as the email address of the device’s creator, Josh Hickman, in the official records. This email was used to register multiple application accounts on the Android 10 device, making it a key point of interest in the forensic analysis.

The second central evidence entity is the name “Josh Hickman”, the creator of the Android 10 device image. As expected, his name appears frequently across various applications on the device, highlighting his key role in the dataset.

Interestingly, the third and fourth central evidence entities, “Christine Teigen” and “chrissyteigen”, were not recorded in the official documentation. Through further investigation, including external Internet searches, we identified Christine Teigen as a well-known American model, actress, author, and television personality. The username “chrissyteigen”, extracted from Twitter, now known as X, data within the device, corresponds to her official Twitter handle. Several tweets mentioning Christine Teigen on the device suggest that the user followed and interacted with content related to her, indicating a potential personal interest or connection.

The identification of these four central nodes demonstrates the effectiveness of our approach in highlighting key evidence entities in digital forensics investigation. By analyzing the FIGs, forensic investigators can prioritize entities that play significant roles, providing valuable leads and insights.

Additionally, a completely separate subgraph containing information from the Telegram application appears in the FIG. Further analysis revealed that some evidence entities in this subgraph, like the usernames “un1” and “un2,” are placeholders in the application. These usernames are dynamically replaced during execution, acting as markers rather than actual user-generated data. Similarly, “Bob Harris” and “Adler Toberg” were identified as default names used within the application for theme or font preview purposes. As a result, these evidence entities do not establish any relationship with other entities on the device. However, their inclusion in the FIG demonstrates the comprehensive coverage of our approach, ensuring that all potential evidence entities, regardless of their contextual relevance, are extracted and considered during forensic analysis.

VI. CHALLENGES AND SOLUTIONS

One of the primary challenges was the variation in file structures and features across different file types. Each file format, such as database files and XML files, presents unique characteristics that affect the extraction and analysis of evidence entities. To address this issue, we conducted an in-depth analysis of the structure of each file type and developed customized preprocessing techniques tailored to their specific

TABLE I
COMPARISON OF CONSTRUCTED FIG IN TERMS OF EVIDENCE ENTITIES AND RELATIONSHIPS

	Ground Truth	Constructed FIG Matched	Coverage	Additional Recovered
Number of Evidence Entities	45	42	93.93%	27
Number of Evidence Relationship	23	20	86.96%	83

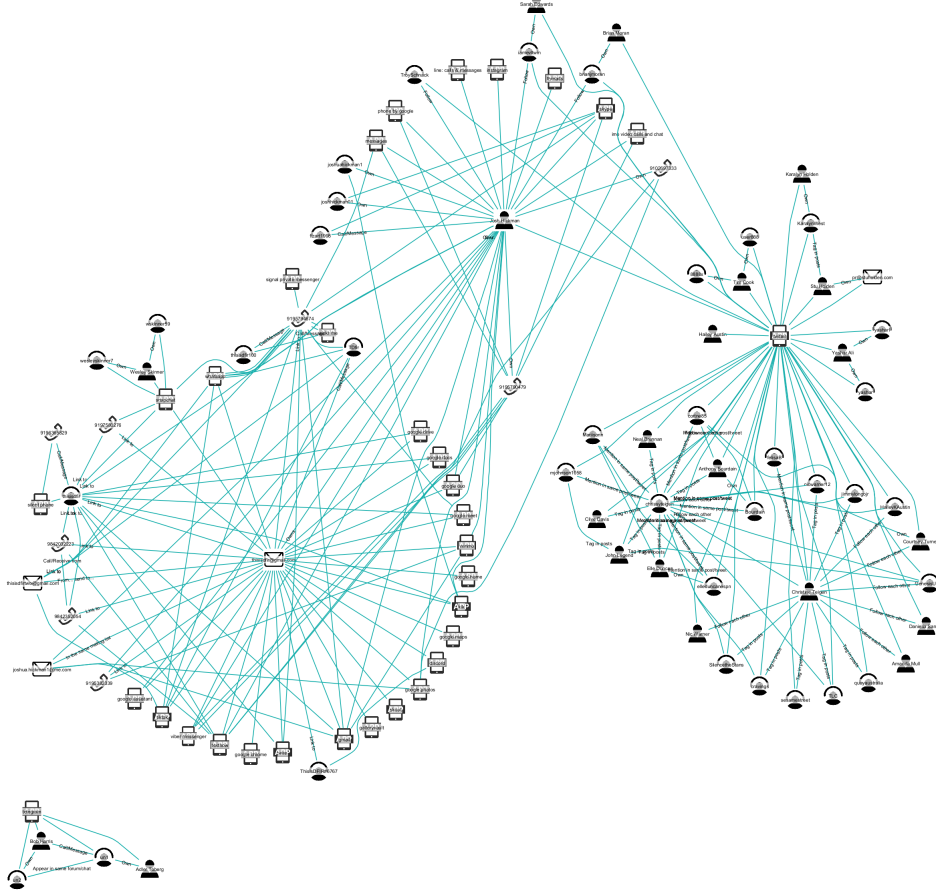


Fig. 3. Experiment Result FIG ²

attributes. This approach ensured that the extracted data was consistent and reliable for further processing by the LLM.

Another significant challenge was the occasional misidentification of evidence entities by the LLM. In some instances, the model extracted entities that did not align with the target evidence types, leading to irrelevant results. To mitigate this issue, we employed prompt engineering techniques by designing constraint-based prompts that precisely define the including and excluding criteria for each evidence entity type. These prompts explicitly specify the characteristics that the target evidence entities should possess and the attributes that should be excluded. This strategy significantly improved the precision of the evidence extraction process.

VII. THREATS TO VALIDITY

The primary threat to the validity of our study is the generalizability of the experimental results across different apps. Each application may store critical user information in varying locations and formats, which can impact the effectiveness of our approach in real-world forensic scenarios. To mitigate this issue, we selected an Android 10 device image populated with real-world user data across several widely used apps as the experiment object. This selection ensures that our approach is tested against diverse application storage structures, enhancing the robustness and applicability of our findings.

Another potential threat to validity is the accuracy of evidence extraction by the LLM. Since LLMs operate on a probabilistic basis, their outputs inherently contain elements of randomness, as they generate responses based on predicted likelihoods within a given context [26]. This variability could

²(Repository for the clear large-scale version of the FIG)
<https://figshare.com/s/1ded2ab80548f8850385>

introduce inconsistencies in the extracted evidence entities and their relationships. To address this challenge, we carefully fine-tune the “temperature” parameter of the LLM, which controls the randomness of the model’s responses, ensuring more deterministic and consistent outputs. Additionally, we focused on optimizing the prompt design to improve the accuracy and reliability of the model’s responses. To further ensure consistency, we conducted multiple iterations of the experiment, verifying that our approach consistently produces the same results with high accuracy.

VIII. CONCLUSION

With the increasing reliance on mobile devices and the vast amounts of digital evidence they generate, effective forensic analysis has become a critical aspect of modern investigations. Traditional forensic methods often struggle to efficiently extract and analyze evidence entities and their relationships due to the diverse data structures and complex storage locations within mobile apps. Thus, we propose an LLM-driven approach to automate the extraction of evidence entities and the construction of FIGs.

Our experimental results demonstrate the effectiveness of our approach, achieving an Evidence Entity Coverage of 93.33%, and an Evidence Relationship Coverage of 86.96%. Furthermore, our method uncovered 27 additional evidence entities and 83 additional relationships beyond those recorded in the official documentation, showcasing its effectiveness in identifying potential forensic artifacts that may have been overlooked in manual investigations.

For future work, we will explore the integration of advanced LLM models, such as fine-tuned versions of gpt-4o, Davinci, and Llama3, to enhance the accuracy and efficiency of evidence extraction and relationship inference across diverse forensic domains. Additionally, we plan to expand both the types and scale of evidence entities represented in the FIG to enhance its clarity, informativeness, and overall utility in digital forensic investigations.

ACKNOWLEDGMENT

This work is supported in part by U.S. Bureau of Justice Assistance (BJA) under 2019-DF-BX-K001.

REFERENCES

- [1] N. Kumar. (2024) Smartphone usage statistics 2024 (worldwide data by country). [Online]. Available: <https://www.demandsage.com/smartphone-usage-statistics/#:~:text=7.21%20billion%20people%20worldwide%20use,time%20in%20the%20United%20States.>
- [2] H. Zhou, P. D. Nguyen, L. Deng, W. Xu, J. Dehlinger, and S. Chakraborty, “Reconstructing android user behavior through timestamped state models,” in *2023 IEEE 47th Annual Computers, Software, and Applications Conference (COMPSAC)*. IEEE, 2023, pp. 574–585.
- [3] H. Zhou, L. Deng, W. Xu, W. Yu, J. Dehlinger, and S. Chakraborty, “Towards internet of things (iot) forensics analysis on intelligent robot vacuum systems,” in *2022 IEEE/ACIS 20th International Conference on Software Engineering Research, Management and Applications (SERA)*. IEEE, 2022, pp. 91–98.
- [4] G. Horsman and N. Sunde, “Unboxing the digital forensic investigation process,” *Science & Justice*, vol. 62, no. 2, pp. 171–180, 2022.
- [5] J. Patel, “Forensic investigation life cycle (file) using 6 ‘r’ policy for digital evidence collection and legal prosecution,” *Int. J. Emerg. Trends Technol.*, vol. 2, no. 1, pp. 129–132, 2013.
- [6] H. Alatawi, K. Alenazi, S. Alshehri, S. Alshamakh, M. Mustafa, and A. Aljaedi, “Mobile forensics: A review,” in *2020 International Conference on Computing and Information Technology (ICCIT-1441)*. IEEE, 2020, pp. 1–6.
- [7] B. Patel and P. S. Mann, “A survey on mobile digital forensic: Taxonomy, tools, and challenges,” *Security and Privacy*, p. e470, 2024.
- [8] M. Chen, J. Tworek, H. Jun, Q. Yuan, H. P. D. O. Pinto, J. Kaplan, H. Edwards, Y. Burda, N. Joseph, G. Brockman *et al.*, “Evaluating large language models trained on code,” *arXiv preprint arXiv:2107.03374*, 2021.
- [9] E. Kasneci, K. Seßler, S. Küchemann, M. Bannert, D. Dementieva, F. Fischer, U. Gasser, G. Groh, S. Günemann, E. Hüllermeier *et al.*, “Chatgpt for good? on opportunities and challenges of large language models for education,” *Learning and individual differences*, vol. 103, p. 102274, 2023.
- [10] Y. Chang, X. Wang, J. Wang, Y. Wu, L. Yang, K. Zhu, H. Chen, X. Yi, C. Wang, Y. Wang *et al.*, “A survey on evaluation of large language models,” *ACM Transactions on Intelligent Systems and Technology*, vol. 15, no. 3, pp. 1–45, 2024.
- [11] H. Zhou, X. Huang, and L. Deng, “Enhancing network traffic classification with large language models,” in *2024 IEEE International Conference on Big Data (BigData)*. IEEE, 2024, pp. 7282–7291.
- [12] Y. Deng, N. Zhao, and X. Huang, “Early chatgpt user portrait through the lens of data,” in *2023 IEEE International Conference on Big Data (BigData)*. IEEE, 2023, pp. 4770–4775.
- [13] W. Xu and D. Xu, “Visualizing and reasoning about presentable digital forensic evidence with knowledge graphs,” in *2022 19th Annual International Conference on Privacy, Security & Trust (PST)*. IEEE, 2022, pp. 1–10.
- [14] N. R. Roy, A. K. Khanna, and L. Aneja, “Android phone forensic: Tools and techniques,” in *2016 International Conference on Computing, Communication and Automation (ICCCA)*. IEEE, 2016, pp. 605–610.
- [15] S. C. Sathe and N. M. Dongre, “Data acquisition techniques in mobile forensics,” in *2018 2nd international conference on inventive systems and control (icisc)*. IEEE, 2018, pp. 280–286.
- [16] H. H. Lwin, W. P. Aung, and K. K. Lin, “Comparative analysis of android mobile forensics tools,” in *2020 IEEE Conference on Computer Applications (ICCA)*. IEEE, 2020, pp. 1–6.
- [17] L. Peng, X. Zhu, and P. Zhang, “A machine learning-based framework for mobile forensics,” in *2020 IEEE 20th International Conference on Communication Technology (ICCT)*. IEEE, 2020, pp. 1551–1555.
- [18] F. Marturana, G. Me, R. Berte, and S. Tacconi, “A quantitative approach to triaging in mobile forensics,” in *2011 IEEE 10th International Conference on Trust, Security and Privacy in Computing and Communications*. IEEE, 2011, pp. 582–588.
- [19] P. Dudhe and S. Gupta, “Machine learning approaches in mobile data forensic: An overview,” in *Proceedings of the Third International Conference on Information Management and Machine Intelligence: ICIMMI 2021*. Springer, 2022, pp. 93–102.
- [20] OpenAI. (2024) Gpt-4o introduction. [Online]. Available: <https://platform.openai.com/docs/models/gpt-4o>
- [21] J. Hickman. (2020) Android 10. [Online]. Available: <https://digitalcorpora.org/corpora/cell-phones/android-10/>
- [22] C. Hummert and D. Pawlaszczyk, *Mobile Forensics—The File Format Handbook: Common File Formats and File Systems Used in Mobile Devices*. Springer Nature, 2022.
- [23] M. Franceschet, D. Gubiani, A. Montanari, and C. Piazza, “From entity relationship to xml schema: a graph-theoretic approach,” in *Database and XML Technologies: 6th International XML Database Symposium, XSym 2009, Lyon, France, August 24, 2009. Proceedings 6*. Springer, 2009, pp. 165–179.
- [24] SQLite. (2024) Wal-mode file format. [Online]. Available: https://www.sqlite.org/walformat.html?utm_source=chatgpt.com
- [25] E. Hyrkas. (2023) How to tune llm parameters for top performance: Understanding temperature, top k, and top p. [Online]. Available: <https://www.phdata.io/blog/how-to-tune-llm-parameters-for-top-performance-understanding-temperature-top-k-and-top-p/#:~:text=What%20is%20the%20Top%20DP,%25%20and%200%20being%200%25.>
- [26] M.-A. Blete. (2023) Llms: Determinism randomness. [Online]. Available: <https://medium.com/@mariealice.blete/llms-determinism-randomness-36d3f3f1f793>